

HOMER, AK, USA

WMO: 703410

Lat: 59.642N

Lon: 151.491W

Elev: 64

StdP: 14.66

Time Zone: -9.00 (NAK)

Period: 94-19

WBAN: 25507

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	0.8	5.0	-10.3	3.2	5.9	-6.6	3.9	8.2	27.0	32.5	24.5	30.6	5.1	40	0.697

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.6	66.1	56.9	63.8	55.9	62.0	54.9	58.3	64.5	57.1	62.5	55.9	60.8	8.8	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
55.1	64.7	60.6	54.1	62.3	59.4	53.0	59.8	58.3	25.2	64.5	24.4	62.6	23.7	60.9	66.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
21.9	18.7	16.7	-2.9	72.0	5.6	3.6	-7.0	74.6	-10.2	76.7	-13.4	78.7	-17.5	81.3		
			-3.7	60.6	5.4	2.1	-7.5	62.2	-10.6	63.4	-13.7	64.6	-17.6	66.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
Temperatures, Degree-Days and Degree-Hours	DBAvg	39.6	25.2	28.2	29.6	38.0	45.3	51.5	55.2	54.6	49.0	40.0	30.6	26.8		
	DBStd	12.94	11.19	9.21	8.17	5.16	4.44	3.71	3.28	3.47	4.35	6.48	9.00	9.53		
	HDD50	4236	768	610	633	359	157	24	2	4	66	312	583	718		
	HDD65	9282	1233	1030	1098	809	612	405	303	323	479	774	1033	1183		
	CDD50	429	0	0	0	0	10	68	165	146	37	3	0	0		
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH74	4	0	0	0	0	0	1	2	1	0	0	0	0		
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0		
	WSAvg	6.3	6.3	6.4	6.8	6.9	7.0	6.6	6.0	5.6	5.8	6.0	6.2	6.7		
	PrecAvg	25.4	2.5	2.0	1.7	1.1	1.1	0.9	1.5	2.4	3.2	3.1	2.7	3.1		
Precipitation	PrecMax	38.1	8.7	5.6	6.0	4.1	2.3	2.5	3.7	5.2	6.2	8.5	8.7	7.8		
	PrecMin	14.3	0.4	0.1	0.0	0.1	0.1	0.0	0.2	0.5	0.8	0.8	0.2	0.1		
	PrecStd	5.3	1.9	1.3	1.3	0.8	0.6	0.6	0.9	1.1	1.4	1.8	2.1	2.0		
	PrecStd	5.3	1.9	1.3	1.3	0.8	0.6	0.6	0.9	1.1	1.4	1.8	2.1	2.0		
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	49.0	46.8	48.7	55.2	63.5	68.7	70.6	69.1	63.1	56.2	51.2	47.3		
		MCWB	42.7	41.3	41.6	45.8	49.2	56.6	59.5	58.5	54.1	49.4	44.9	42.4		
	2%	DB	44.8	43.2	45.1	50.9	58.7	63.7	66.0	65.8	60.3	53.6	48.1	44.6		
		MCWB	40.2	39.0	40.1	42.7	48.5	54.4	57.5	57.2	52.8	47.3	43.6	40.2		
	5%	DB	41.9	40.9	42.5	48.6	55.6	61.1	63.8	63.4	57.9	51.5	45.5	42.4		
		MCWB	38.1	37.2	38.0	41.7	47.4	52.9	56.5	56.4	51.8	46.2	41.9	38.7		
	10%	DB	39.6	39.1	40.2	46.5	53.5	58.9	62.1	61.4	56.3	49.6	43.0	39.9		
		MCWB	36.8	35.9	36.2	40.5	46.4	51.7	55.6	55.3	51.0	45.1	40.1	37.0		
	0.4%	WB	43.0	42.1	42.6	46.9	51.6	58.1	60.1	60.3	55.8	50.7	46.1	43.2		
		MCDB	48.4	45.9	47.3	54.5	59.4	66.5	68.3	66.4	60.6	55.0	49.6	46.8		
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	40.5	39.6	40.7	43.9	49.7	55.5	58.4	58.5	54.1	48.4	44.1	40.7		
		MCDB	44.3	42.7	44.7	49.2	56.8	62.0	64.8	64.1	58.1	52.3	47.4	44.0		
	5%	WB	38.6	37.6	38.6	42.3	48.2	53.8	57.2	57.2	52.9	47.0	42.3	38.9		
		MCDB	41.6	40.1	42.1	47.4	54.9	59.6	62.7	62.1	56.6	50.5	45.3	41.9		
	10%	WB	36.9	36.2	36.7	41.1	46.8	52.4	56.1	56.0	51.8	45.8	40.3	37.1		
		MCDB	39.4	38.7	39.8	46.0	52.8	57.7	61.2	60.5	55.3	49.3	43.0	39.7		
Mean Daily Temperature Range	5% DB	MDBR	10.8	11.1	12.1	12.8	13.3	12.6	11.6	12.3	12.4	11.6	10.9	10.5		
		MCDBR	10.3	10.2	12.3	15.7	17.6	17.0	14.9	15.6	14.3	10.9	10.0	10.1		
		MCWBR	7.6	7.5	8.7	10.0	10.4	9.5	8.4	9.1	9.2	7.4	7.5	7.8		
		MCDBR	10.0	9.8	10.8	14.4	16.5	15.5	13.6	13.9	12.3	9.9	9.1	9.6		
	5% WB	MCWBR	7.7	7.6	8.0	9.6	10.1	9.1	8.0	8.6	8.4	7.1	7.0	7.8		
		taub	0.254	0.302	0.309	0.352	0.370	0.358	0.364	0.356	0.333	0.314	0.272	0.258		
		taud	2.207	2.263	2.335	2.283	2.309	2.428	2.454	2.480	2.514	2.409	2.166	2.005		
		Ebn at Noon	174	223	263	269	272	278	273	266	255	219	157	110		
All-Sky Solar Radiation	RadAvg	113	307	735	1171	1559	1695	1509	1199	779	384	157	65			
	RadStd	14	48	91	149	131	118	122	147	116	40	27	8			

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	+3.78	+3.42	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (18 neighbors)	+1.33	+4.01	+3.73	N/A	N/A	N/A	N/A	-452	N/A	N/A

Nomenclature: See separate page